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Space-weather Utilities for Research and Forecasting (SURF): A tool for investigating hydrodynamic aspects of solar wind and coronal mass ejection expansion and evolution.

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Abstract We introduce Space-weather Utilities for Research and Forecasting (SURF), a flexible modelling framework for reconstructing and forecasting large-scale solar wind structure. SURF incorporates both the incompressible "HUXt" model and a newly developed one-dimensional, compressible hydrodynamic solver, "hydro.". The new solver relaxes the incompressibility assumption while retaining the computational efficiency required for ensemble forecasting, uncertainty quantification, and large parametric studies. SURF-hydro solves the 1D Euler equations in spherical geometry using finite-volume methods with Riemann solvers and second-order spatial reconstruction. It is shown to accurately reproduce an analytical solution to pressure-driven expansion of a uniform spherical solar wind, demonstrating adequate numerical fidelity. Using 30 years of near-Earth OMNI observations, we derive empirical relations between speed, density, and temperature at 1 AU and map these back to 0.1 AU to provide non-equilibrium inner-boundary conditions for SURF-hydro (and for use with other solar wind models). This approach produces more realistic solar wind speed, density and temperature values and variability at 1 AU than simulations based on equilibrium thermodynamic relations at 0.1 AU. SURF-hydro is also shown to reproduce the key features of coronal mass ejection (CME) expansion and evolution, enabling rapid exploration of CME parameter space. For one particular realisation of a structured solar wind, sensitivity tests demonstrate that CME density and temperature at 0.1 AU alter 1 AU transit times and arrival speeds by 15–20%, highlighting an under-explored source of forecast uncertainty. SURF-hydro therefore provides a computationally efficient bridge between idealised models and full 3D magnetohydrodynamic simulations, enabling systematic investigation of boundary condition assumptions and CME parameter sensitivity for both research and operational space-weather forecasting.

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1. Introduction

Reconstruction of the state of the large-scale solar wind underpins a huge range of scientific and space weather forecasting applications. The standard methodology is to couple coronal and solar wind (or heliospheric) models, typically with an interface around 0.1 AU, where the solar wind flow is already super-magnetosonic. There are now a wide range of three-dimensional magnetohydrodyanmic (3DMHD) solar wind models, such as Enlil (Odstrcil and Pizzo 1999), EUHFORIA (Pomoell and Poedts 2018), SWMF (Toth et al. 2005), SUSANOO (Shiota and Kataoka 2016), HelioMAS (Riley, Linker, and Mikic 2001), ICARUS (Baratashvili et al. 2025) and GAMERA (Merkin et al. 2016). These models accurately capture the large-scale dynamics of a magnetised fluid flow. But there are many situations where the computational expense and code complexity of 3DMHD becomes prohibitive. These include large ensembles of runs for uncertainty quantification (Owens and Riley 2017) and data assimilation (Barnard et al. 2023), large-scale parametric studies (Gyeltshen et al. 2026), long-duration runs for planetary studies (O'Donoghue et al. 2025), development of adjoint models for data assimilation (Lang and Owens 2019), etc. In such situations it is useful to sacrifice a small amount of physical fidelity for greatly increased computational efficiency and code simplicity. Highly idealised and empirical models can be both extremely rapid and useful, particularly for specifc applications in forecasting (e.g. Hinterreiter et al. 2021). But there is also a middle ground, using discretised numerical solutions with what is best referred to as a "reduced-physics" approximation. The heliospheric upwind extrapolation model with time dependence (HUXt; Owens et al. 2020; Barnard and Owens 2022) is one such reduced-physics model that has found use for both forecasting (https://swxforecastlab.org/forecasts.html) and a diverse range of scientific applications (e.g. O'Donoghue et al. 2025; Watson et al. 2025; Chi et al. 2023; Yu et al. 2023).

HUXt treats the solar wind as a one-dimensional, incompressible, hydrodyanmic flow. The 1D approximation is well justified as the solar wind flow is observed to be highly radial in most situations (e.g. Scherer et al. 2001; Owens and Cargill 2004), and even complex three dimensional structures like coronal mass ejections (CMEs) behave as quasi-one-dimensional structures – at least to first order – owing to their rapid spherical expansion (Owens, Lockwood, and Barnard 2017). The hydrodyanmic approximation can also be justified with simple order-of-magnitude arguments, with the 1-AU solar wind displaying dynamic flow pressure around two orders of magnitude higher than either the thermal or magnetic pressures in most circumstances. The incompressible assumption is perhaps the hardest to justify. While HUXt has been shown to produce flow fields in very close agreement (within 5%) with 3DMHD for the same boundary conditions (Owens et al. 2020), we here seek to relax the incompressible approximation. To this end, we introduce a compressible hydrodyanmic solver – "hydro" – that does not sacrifice too much of the computationally efficiency that makes HUXt a useful forecast and research tool.

A significant amount of open-source supporting infrastructure has been built up around the HUXt model. The HUXt codebase now contains easy-to-use capability for driving HUXt with semi-empirical (Arge et al. 2003) and full MHD (Linker et al. 1999) coronal model output, as well as from coronal tomography (Bunting et al. 2024) and cone-model CMEs (Zhao, Plunkett, and Liu 2002), and boundary conditions derived directly from in situ observations (Owens et al. 2026). Support exists for both inner heliosphere ensemble (Owens and Riley 2017) and outer heliosphere science (O'Donoghue et al. 2025), and for both steady-state and time-dependent (Owens, Barnard, and Arge 2024) ambient solar wind modelling. HUXt can readily be run in pseudo 2- or 3-dimensions, with field-line tracing, time series and CME arrival times at any specified body, and with an array of visualisation options. We seek to exploit this pre-existing modelling framework for the new hydro model and other future developments. For this reason, we refer to the general modelling framework as "SURF"; Space-weather Utilities for Research and Forecasting. HUXt and hydro are modelling options within SURF. This framework also permits future developments to be easily incorporated, such as one-dimensional MHD solvers (e.g. Tao et al. 2005; Riley and Ben-Nun 2022), or true two- or three-dimensional hydrodynamic solvers.

Section 2 reviews the physical equations we seek to solve with the new 'SURF-hydro' model, while Section 3 describes the specific numerical implementation used. Section 4 outlines the analytical 'Parker nozzle' model of a spherically expanding fluid which we use to validate the numerical SURF-hydro solutions. The results are split across two sections: Section 5 details how the ambient solar wind boundary conditions are determined and the resulting model solutions at 1 AU, whereas Section 6. Section 7 summaries the results and possible future avenues for research.

2. Physical basis

The evolution of the solar wind flow can be approximated by the equations of mass, momentum and energy conservation for a hydrodynamic flow. We here follow the basic framework outlined in the 'pyro' python library of hydrodynamics examples (Harpole et al. 2019; Zingale 2014). However, we develop a stand-alone solver, fully integrated within the SURF framework, rather than produce a 'wrapper' for the existing pyro code.

Ignoring gravity and viscous stress, we seek to solve:

$$\text{conservation of mass: } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \tag{1}$$

$$\text{conservation of momentum: } \frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) + \nabla p = 0, \tag{2}$$

$$\text{and conservation of energy: } \frac{\partial (\rho E)}{\partial t} + \nabla \cdot (\mathbf{v} E) + \nabla \cdot (p \mathbf{v}) = 0, \tag{3}$$

where ρ is the mass density, $\mathbf{v}$ is the fluid velocity, p is the fluid pressure and E is the total fluid energy density, given by:

$$E = \frac{1}{2}\rho\mathbf{v}^2 + E_{INT}, \tag{4}$$

where E_{INT} is the internal energy density of the gas. For an ideal gas:

$$E_{INT} = \frac{p}{\gamma - 1}, \tag{5}$$

where γ is the polytropic index, taken to be 1.5.

Considering only the radial (r) direction in spherical geometry, $\mathbf{v}$ is reduced to the radial speed, v, and the 1D Euler equation becomes:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{1}{r^2}\frac{\partial (r^2\mathbf{F})}{\partial r} = \mathbf{S}, \tag{6}$$

where the conserved variables, $\mathbf{U}$, are:

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho v \\ \rho E \end{bmatrix}, \tag{7}$$

and the fluxes, $\mathbf{F}$, are:

$$\mathbf{F} = \begin{bmatrix} \rho v \\ \rho v^2 + p \\ v(E+p) \end{bmatrix}. \tag{8}$$

$\mathbf{S}$ is the geometric source vector. In Cartesian coordinates, $\mathbf{S} = 0$. However, in spherical coordinates, there is a geometric source of momentum arising from the fact that the surface area of a sphere grows as $1/r^2$ (Harpole et al. 2019; Zingale 2014). This results in a geometric pressure gradient. Therefore:

$$\mathbf{S} = \begin{bmatrix} 0 \\ \frac{2p}{r} \\ 0 \end{bmatrix}. \tag{9}$$

3. Numerical implementation

In order to solve the physical equations in Section 2, they are discretized in radial distance, r, and time, t, using a finite volume method. Thus a conserved variable at coordinates (r, t) is written as:

$$\mathbf{U}(r,t) \rightarrow \mathbf{U}_i^n = \mathbf{U}(r_i, t_n) \tag{10}$$

where r_i is the radial distance of the centre of the i^{th} grid cell, and t_n is the n^{th} time step. We solve on a fixed, uniform radial grid, with grid cell spacings of Δr. We refer to the interface between cell i and $i+1$ as $i+1/2$. Time steps are denoted by n, with spacing between consecutive time steps as Δt. This time step is variable to ensure stability (see discussion below). Note that the SURF-hydro output, however, is linearly interpolated onto a fixed, uniform time grid to ensure compatibility with the existing plotting and analysis codebase.

As each grid cell is part of a spherical shell, the area at each cell interface is $A_{i\pm1/2} = 4\pi r_{i\pm1/2}^2$. The evolution of a conserved quantity at grid cell i is then given by:

$$\frac{d\mathbf{U}}{dt} = \mathcal{L}(\mathbf{U}) \tag{11}$$

where $\mathcal{L}$ is the spatial discretization operator:

$$\mathcal{L}(\mathbf{U}) = -\frac{1}{V_i}\left(r_{i+1/2}^2\mathbf{F}_{i+1/2} - r_{i-1/2}^2\mathbf{F}_{i-1/2}\right) + \mathbf{S}_i \tag{12}$$

and V_i is the volume of the shell at distance r_i:

$$V_i = \frac{1}{3}(r_{i+1/2}^3 - r_{i-1/2}^3). \tag{13}$$

As shown in Equation 9, the geometric source terms are zero for mass and energy. For momentum, however, spherical geometry results in a net component of the pressure force acting in the radial direction. Thus the source term is given by:

$$\mathbf{S_i} = \begin{bmatrix} 0 \\ p_i\frac{A_{i+1/2}-A_{i-1/2}}{v_i} \\ 0 \end{bmatrix} \tag{14}$$

In order to compute the fluxes through the inner and outer cell boundaries, we must first must compute $\mathbf{U}$ either side of the cell interfaces, rather than just the average values at the cell centres. Two reconstruction approaches are used. Like the original HUXt code, one method is a first-order approach, the piecewise constant method (PCM). In PCM, the value either side of the interface is simply assumed to be the value at the cell centre. So at interface $i+1/2$ (the outer boundary of cell i and inner boundary of cell $i+1$) the state at the inner edge of the interface is given by $\mathbf{U_{INNER}} = \mathbf{U_i}$, while the outer edge is given by $\mathbf{U_{OUTER}} = \mathbf{U_{i+1}}$.

However, the default method in SURF-hydro is to use a second-order solution, the Piecewise Linear Method (PLM). This uses the linear gradient in $\mathbf{U}$, such that at the $i+1/2$ interface, $\mathbf{U_{INNER}} = \mathbf{U_{i+1/2}}\Delta\mathbf{U_i}$ and $\mathbf{U_{OUTER}} = \mathbf{U_{i+1}} - 1/2\Delta\mathbf{U}_{i+1}$. This retains sharper features, such as shocks, but requires a gradient limiter. We use the monotonized central limiter (Van Leer 1977) to prevent spurious oscillations while maintaining second-order accuracy away from extrema.

Next, for both reconstruction methods, we must solve for the discontinuities across cells. We use a Harten-Lax-van Leer-Contact solver (HLLC Toro 2009; Toro, Spruce, and Speares 1994) that assumes the solution consists of two waves (inner and outer) and a contact discontinuity in between. Either side of the contact discontinuity – and between the inner and outer call interfaces – are two intermediate 'star states', $\mathbf{U_{*INNER}}$ and $\mathbf{U_{*OUTER}}$. The pressure and velocity are equal in these two regions, but the density and energy may be different. The HLLC solver evaluates the wave speeds and determines which region ($\mathbf{U_{i-1/2}}$, $\mathbf{U_{*INNER}}$, $\mathbf{U_{*OUTER}}$, $\mathbf{U_{i+1/2}}$) is sampled at the interface, then returns the corresponding flux.

The final step is the time update. For the PCM reconstruction, we use the first-order forward Euler time integration, where:

$$\mathbf{U^{n+1}} = \mathbf{U^{n}} + \Delta t \mathcal{L}(\mathbf{U^{n}}) \tag{15}$$

For the PLM reconstruction, it is necessary to use the second-order Runge-Kutta method. Here we first compute the predictor state, $\mathbf{U^{p}}$, which is the same as the first-order solution:

$$\mathbf{U^{p}} = \mathbf{U^{n}} + \Delta t \mathcal{L}(\mathbf{U^{n}}) \tag{16}$$

and then to compute the second-order correction:

$$\mathbf{U^{n+1}} = \mathbf{U^{n}} + \frac{\Delta t}{2} \left[\mathcal{L}(\mathbf{U^{n}}) + \mathcal{L}(\mathbf{U^{p}}) \right] \tag{17}$$

For both PCM and PCL, the time step must be chosen to meet the Courant-Friedrichs-Lewy (CFL) stability condition, such that:

$$\Delta t = C \min_{i} \left(\frac{\Delta r}{|v_i| + c_i} \right) \tag{18}$$

where $c_i = \sqrt{\gamma p_i / \rho_i}$ is the local sound speed and C is a constant to ensure stability. To balance computational efficiency and stability we use default values of $C = 0.8$ for PCM and 0.6 for PLM.

As in HUXt, the user can specify the radial grid, with the time step being automatically adjusted to match the associated CFL conditions. A default grid size of 1.5 solar radii (r_S) is used. Two- and three dimensional HUXt runs involve running a series of independent SURF-HUXt and SURF-hydro one-dimensional models. Thus while longitudinal and latitudinal grid spacing can also be specified, this has no direct numerical or stability implications. By default, 128 longitudinal grid cells are used, and 45 latitudinal cells. All plots and analysis in this study use the default grids.

4. Validation

In order to test the numerical implementations of SURF-hydro described above, we compare with an analytical solution for a steady-state solar wind expansion in diverging spherical geometry. This differs from the classic Parker solar wind solution (Parker 1958) in that it ignores gravity and allows temperature to vary adiabatically. Thus it involves the exact same physics as SURF-hydro.

This solution treats the solar wind acceleration as a one-dimensional isentropic flow through a nozzle with cross-sectional area $A(r) \propto r^2$. It can then be shown (Anderson 2021) that the temperature, density and flow speed are given by: .

$$T(r) = T_0 \left[\frac{1 + \frac{\gamma - 1}{2} M_0^2}{1 + \frac{\gamma - 1}{2} M(r)^2} \right] \tag{19}$$

$$v(r) = v_0 \frac{M(r)}{M_0} \sqrt{\frac{T(r)}{T_0}} \tag{20}$$

$$\rho(r) = \rho_0 \frac{M_0}{M(r)} \sqrt{\frac{T_0}{T(r)}} \left(\frac{r_0}{r} \right)^2 \tag{21}$$

where $M(r) = v(r)/c(r)$ is the local Mach number.

This analytical solution exactly conserves mass flux ($\rho v r^2 = \text{const}$) and provides a benchmark for validating SURF-hydro in smooth, shock-free flows (Toro 2009).

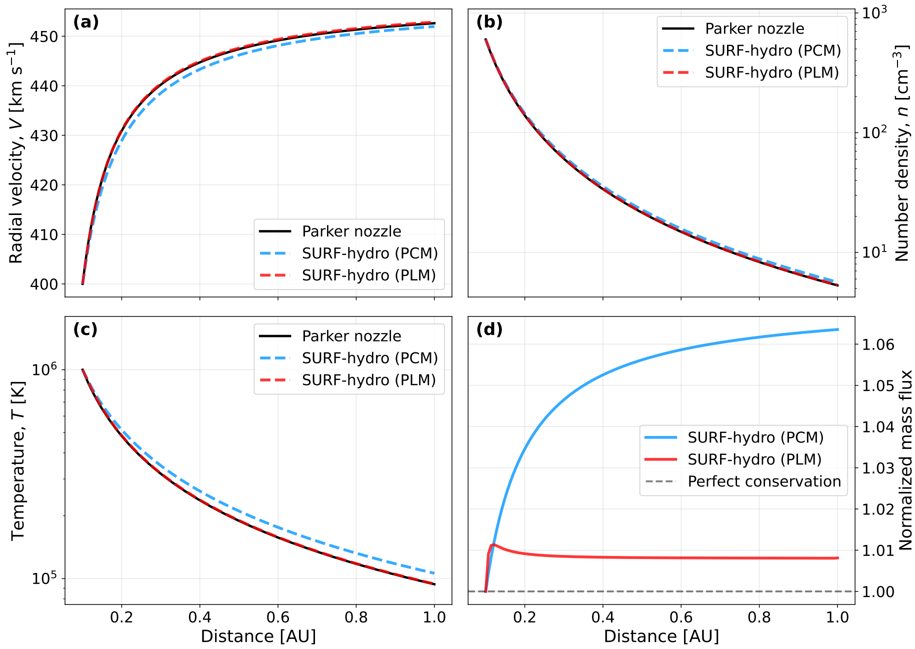

Figure 1. Comparison of SURF-hydro and analytical solar wind solutions from 0.1 to 1 AU (21.5 to 215 Rs). (a) Solar wind speed, v, as a function of heliocentric distance. The analytical solution is shown as a solid black line, with SURF-hydro PCM and PLM solutions shown as blue and red dashed lines, respectively. (b) number density, n_P, assuming a proton plasma, (c) temperature, T (d) fractional change in total mass.

Figure 1 shows comparisons of the SURF-hydro solutions with the analytical steady-state solar wind solution. All models use inner boundaries at 0.1 AU and inner boundary conditions of v = 400 km s^{-1}, T = 1 MK and a mass density computed from assuming a proton plasma of number density $n_P = 600$ cm^{-3}. For the analytical model, these values result in 1 AU properties of $v = 453$ km s^{-1}, T = 94000 K and n_P = 5 cm^{-3}, which are fairly typical values (e.g. Owens 2020). For the SURF-hydro solutions, the model solutions are shown after the standard spin-up time of 5 days, to allow propagation of the inner boundary conditions through the whole 1-AU domain.

By eye, the SURF-hydro-PLM solution to v is nearly identical to the analytical benchmark solution. Over the whole domain, the error in v is 0.04%, in n_P is 0.89%, and in T is 0.85%. For SURF-hydro-PCM, the deviation from the analytical benchmark solution is visible in the plots, particularly for temperature. Specifically, over the whole domain, the error in v is 0.40%, in n_P is 5.63% and in T is 11.17%. Perhaps more important is the conservation of mass in the simulation. For SURF-hydro-PCM, the mass flux deviates from the inner boundary condition by 5.5% at 1 AU and is still continuing to gradually rise even at the outer boundary. This could be reduced by increasing the spatial and temporal resolution of HUXt, but that would come with additional computation cost and PCM would still have difficulty with capturing shocks. By contrast, the second-order scheme of SURF-hydro-PLM conserves mass to within 1%. Furthermore, even this small deviation is due entirely to the second-order scheme needing to interpolate across the inner boundary. Within the main computational domain, mass is conserved to machine accuracy.

For PCM, the 5-day run for a single longitude takes 0.03 seconds on a standard desktop CPU, while the PLM run takes 0.10 seconds. This compares with the original (incompressible) HUXt runtime of the same setup of around 0.02 seconds. Thus the second-order solver does add significant – approximately x5 – computational overhead compared to HUXt, but SURF-hydro is still rapid enough for most of the original HUXt use cases. The PLM solver is used through the remainder of this paper. However, PCM is still included as an option within SURF for specific situations where large numbers of ambient solar wind solutions are required, and the computation time is particularly critical.

5. Ambient solar wind

The solar wind acceleration is prescribed as an analytical function in SURF-HUXt. This empirical relation is intended to mimic the spherical pressure gradient (Riley and Lionello 2011). For SURF-hydro, this process arises naturally, meaning solar wind acceleration is sensitive to the choice of temperature and (to a lesser extent) density at the inner boundary.

Figure 2 uses SURF-hydro to show how v at 1 AU varies for different T and n_P at 0.1 AU. A uniform v at 0.1 AU of 400 km s^{-1} is used throughout. T at 0.1 AU has a strong effect on the resulting solar wind acceleration. A temperature of 1 MK at 0.1 AU produces about a 10% increase in solar wind speed by 1 AU, whilst a 3 MK corona results in around a 30% increase. Conversely, solar wind density

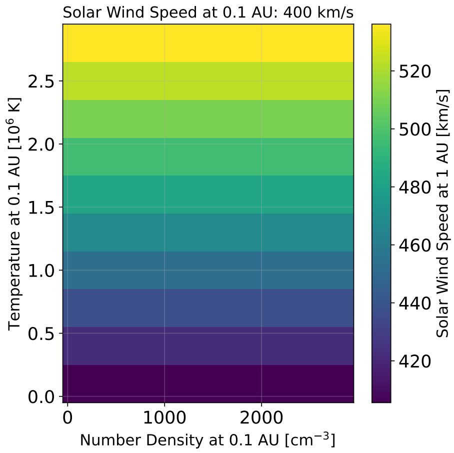

Figure 2. Solar wind speed at 1 AU for a range of proton density and temperatures at 0.1 AU. A uniform solar wind speed at 0.1 AU of 400 km s^{-1} is used throughout.

at 0.1 AU does not affect solar wind acceleration, despite contributing to the solar wind thermal pressure term. This is because the momentum is affected by pressure gradient with radius, which is unaffected by a change in the magnitude of the density. This is also evident from the Parker nozzle solution too, wherein the solar wind speed at a given difference depends only on the initial temperature and initial speed, not the density. Note, however, that once there is structure in the solar wind speed at 0.1 AU, the density will begin to affect dynamics, as it can then produce radial pressure gradients. This is shown in Section 6.

It is therefore important to accurately determine not only speed at 0.1 AU, but also temperature and density. While full MHD solutions of the corona can in principle directly provide this information (Perri et al. 2023), in practice these values often show insufficient variability and the coronal-model output used by solar wind models is typically limited to the radial magnetic field and radial flow speed (Linker et al. 2016; Riley, Linker, and Mikic 2001). Thus it is necessary to estimate n_P and T at the inner boundary on the basis of the inner-boundary v. Typically some form of equilibrium state is assumed at the solar wind model inner boundary. A range of approaches have been used; constant mass flux (Detman et al. 2006), constant kinetic energy flux (Odstrcil and Pizzo 1999; Pomoell and Poedts 2018) or constant momentum flux (Riley et al. 2011) are invoked to infer n_P from v. Then constant thermal pressure is used to determine T. For use with interplanetary scintillation (IPS) derived solar wind speeds, there have been attempts to use purely empirical relations between n_P and T with v, which do not explicitly impose flux or pressure balance (Hayashi, Tokumaru, and Fujiki 2016). However, these are typically applied much further out into the heliosphere ($\sim 50 - 60$ r_S), where most solar wind acceleration has already ceased (e.g. see Figure 1a). Shiota et al. (2014) also used empirical relations derived from *Helios* observations to define n_P and T at the inner boundary, and produced reasonable speeds in near-Earth space. However, they did not consider the accuracy of the resulting density and temperature.

In general, there has not been systematic validation of these assumptions across a large dataset, largely due to the computational cost of such endeavours. It is also likely (see analysis below) that such approaches have been primarily tuned to match solar wind speed in near-Earth space, rather then density and temperature, as that is the more immediate space-weather concern.

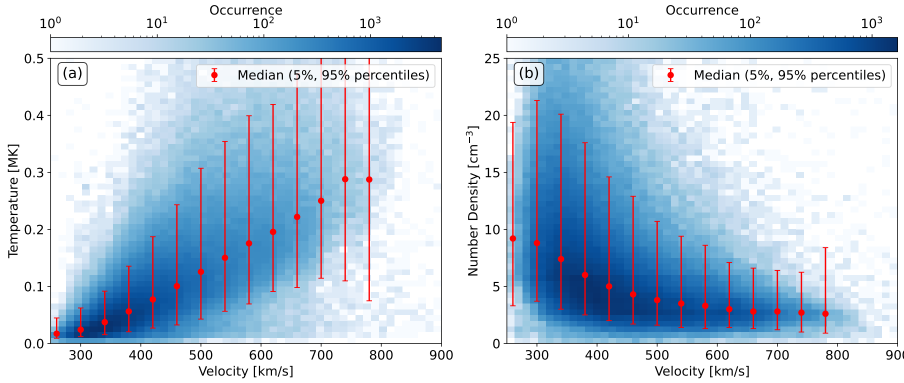

Figure 3. OMNI observations of solar wind parameters at 1 AU over the period 1995-2026 at 1-hour resolution. ICMEs have been removed. Panels show: (a) proton temperature, T, against proton radial velocity, v; (b) Proton number density, n_P, against v. The colour scale shows 2D histograms of occurrence on a log scale. The red dots show the median values in uniform v bins, with red lines spanning the 5- to 95 percentiles.

In future work, SURF-hydro will be used to assess the implications of all the equilibrium approaches over an extended time interval. Here, however, we demonstrate a simple non-equilibrium approach that performs well in reproducing solar wind speed, density and temperature at 1 AU.

Figure 3 shows an analysis of near-Earth solar wind data in order to characterise the relations of density and temperature with radial speed. Using OMNI data for the period of near-contiguous observations (1995 to 2026) at 1-hour resolution, we first remove all periods identified as interplanetary CMEs (ICMEs) using the updated list from Richardson and Cane (2010). Two dimensional occurrence histograms are shown in Figures 3a and b. The data are then binned in v and the median values of density and temperature computed. These are listed in Table 1. However, these are values at 1 AU, but we require relations between n_P, T and v at the SURF-hydro inner boundary radius (typically 0.1 AU).

Each set of v, n_P and T values at 1 AU shown in Table 1 is used with Equations 19 to 21 to compute the 0.1 AU equivalents. These back-mapped values are shown in Table 2 as $v_{0.1\mathrm{AU}}$, $n_{0.1\mathrm{AU}}$ and $T_{0.1\mathrm{AU}}$. Linear interpolation of this look-up table is used to compute n_P and T at the inner boundary. This look-up table approach avoids loss of fidelity that would result from fitting an analytical function to the data and allows for easy updates to the relations resulting from more sophisticated analysis of the observational data sets in the future.

Figure 4a shows the relation between 1-AU and 0.1-AU speeds, calculated assuming the observed 1-AU relations between speed with density and temperature, and applying the analytical Parker solution. Owing to the non-linear

Table 1. Table of median density and temperature for given solar wind speeds at 1 AU based on the OMNI dataset.

$v_{1\mathrm{AU}}$ (km/s)	$n_{1\mathrm{AU}}$ (cm^{-3})	$T_{1\mathrm{AU}}$
220	9.90	9.37e+03
260	9.20	1.71e+04
300	8.80	2.41e+04
340	7.40	3.73e+04
380	6.00	5.61e+04
420	5.00	7.71e+04
460	4.30	1.00e+05
500	3.80	1.26e+05
540	3.50	1.50e+05
580	3.30	1.75e+05
620	3.00	1.96e+05
660	2.80	2.22e+05
700	2.80	2.50e+05
740	2.70	2.88e+05
780	2.60	2.87e+05

relations with density and temperature, the adjustment to speed is also nonlinear, ranging from below 5% at low speeds, to around 15% at higher speeds. We suggest that this is the correction that should be applied to WSA coronal model speeds for use as 0.1 AU inner boundary conditions with dynamical solar wind models such as Enlil, ICARUS and EUHFORIA.

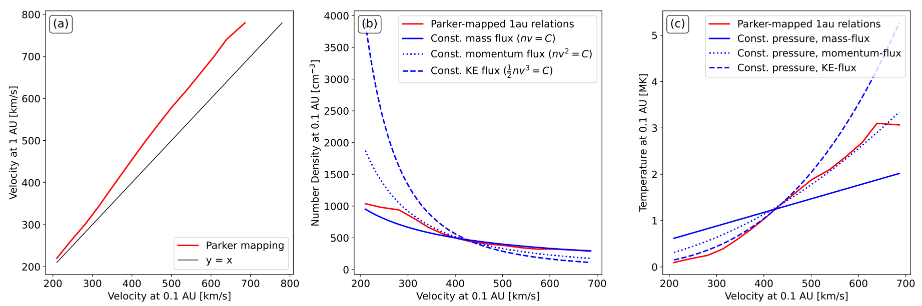

Figure 4. Relations used to construct the inner boundary conditions in SURF-hydro. (a) The red line shows relation between solar wind speed at 1 AU – such as is provided by the WSA model – and 0.1 AU, assuming the 1-AU relations with density and temperature, and using the Parker nozzle solution to solar wind acceleration. (b) Relations between solar wind speed and density at 0.1 AU. The SURF-hydro method – based on observed 1-AU relations – is shown in red. For comparison, assumptions of constant mass flux and constant kinetic energy flux are shown by solid and dashed blue lines, respectively. (c) Relations between speed and temperature at 0.1 AU. The SURF-hydro method is shown in red. Solid and dashed blue lines shown the assumption of constant thermal pressure, using density from constant mass flux and constant kinetic energy flux, respectively.

Figures 4b and c show the relation of speed to density and temperature, respectively, at 0.1 AU. The SURF-hydro relations shown in red are based on the observed 1-AU relations from OMNI data, backmapped using the Parker nozzle solution. Other commonly used assumptions of constant mass flux, constant momentum flux and constant kinetic energy flux are shown for comparison (constants are chosen to match the SURF densities and temperatures at around 400 km/s). Compared to the backmapped OMNI relations, constant kinetic energy flux tends to produce too much variation in density and temperature. Constant momentum flux produces too high density at low speeds, but broadly similar variations in temperature (though with approximately a factor 2 overestimate at low speeds). Conversely, assuming constant mass flux produces similar trends in density, but underestimates the temperature variation. The backmapped OMNI relations thus offer a means to avoid introducing biases associated with all of the equilibrium boundary condition assumptions.

Computation of values at 0.1 AU is shown purely as a typical example. In practice, we use the relations at 1 AU (Table 1) to compute look-up tables at the specific inner boundary distance required for each SURF-hydro run, which varies for different observational input sources and use cases. As with many of the boundary condition construction methods, these relations could be used with any solar wind model, not just the SURF models.

Table 2. Table of solar wind speeds at 1 AU, with equivalent speeds and median densities and temperatures at 0.1 AU. These are based on backmapping the values in Table 1 using the analytical functions in Equations 19 to 21.

$v_{1\mathrm{AU}}$ (km/s)	$v_{0.1\mathrm{AU}}$ (km/s)	$n_{0.1\mathrm{AU}}$ (cm^{-3})	$T_{0.1\mathrm{AU}}$ (K)
220	210	1037.0	9.59e+04
260	244	978.7	1.76e+05
300	281	940.1	2.49e+05
340	313	802.9	3.89e+05
380	343	663.9	5.91e+05
420	374	561.8	8.17e+05
460	404	489.1	1.07e+06
500	435	436.4	1.35e+06
540	468	403.7	1.61e+06
580	502	381.5	1.89e+06
620	539	345.3	2.10e+06
660	573	322.3	2.38e+06
700	608	322.5	2.68e+06
740	639	312.7	3.10e+06
780	686	295.6	3.06e+06

In order to demonstrate how well these empirical relations work, we produce an example solar wind hindcast with SURF-hydro. OMNI near-Earth solar wind observations for 2019-8-18 to 2019-9-15 are first used to construct v as a function of Carrington longitude at 0.1 AU, following the methodology of Owens et al. (2026). Estimates of n_P and T at 0.1 AU are then derived from the empirical relations in Table 2. The resulting 0.1 AU boundary conditions are shown in Figure 5.

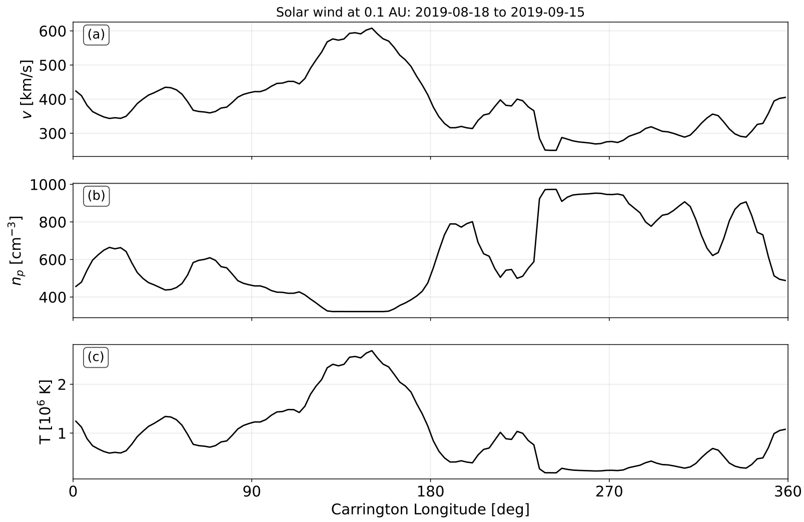

Figure 5. Solar wind properties at 0.1 AU as a function of Carrington longitude. (a) Solar wind speed at 0.1 AU, $v_{0.1\mathrm{AU}}$, generated by 'backmapping' near-Earth OMNI solar wind observations for the period 2019-8-18 to 2019-9-15, following the methodology of Owens et al. (2026). (b) Proton number density at 0.1 AU derived from $v_{0.1\mathrm{AU}}$ and the relations in Table 2 (c) Proton temperature at 0.1 AU, derived from $v_{0.1\mathrm{AU}}$ and the relations in Table 2.

The 0.1 AU boundary conditions were then used to drive both SURF-HUXt and SURF-hydro forward in time for a further 27 days from 2019-9-10. The results are shown in Figure 6. Note that only solar wind speed observations from prior to the period shown in Figure 6 were used to make the hindcast. And no density or temperature observations were used at all.

The SURF hindcast reproduces the general solar wind speed structure in near-Earth space very well, as the previous 27 days had also been similar. Of course, this is not always the case; this interval was specifically chosen as it is a representative example of recurrent solar wind and we here seek to determine how well density and temperature can be reconstructed when the large-scale solar wind speed structure is broadly correct. The results are extremely promising. The base-level proton number density of around 5 cm^{-3} is well captured, with stream interaction regions (SIRs) producing solar wind compressions and density spikes lasting 10s of hours with magnitudes of 10s cm^{-3}. There is a similar picture for temperature, with both observed and modelled proton temperature ranging from around 5×10^4 K in very slow wind, to 5×10^5 K in the compressed solar wind at stream interaction regions. For the major stream interaction region on 2019-9-27, SURF-hydro overestimates the temperature and density magnitudes

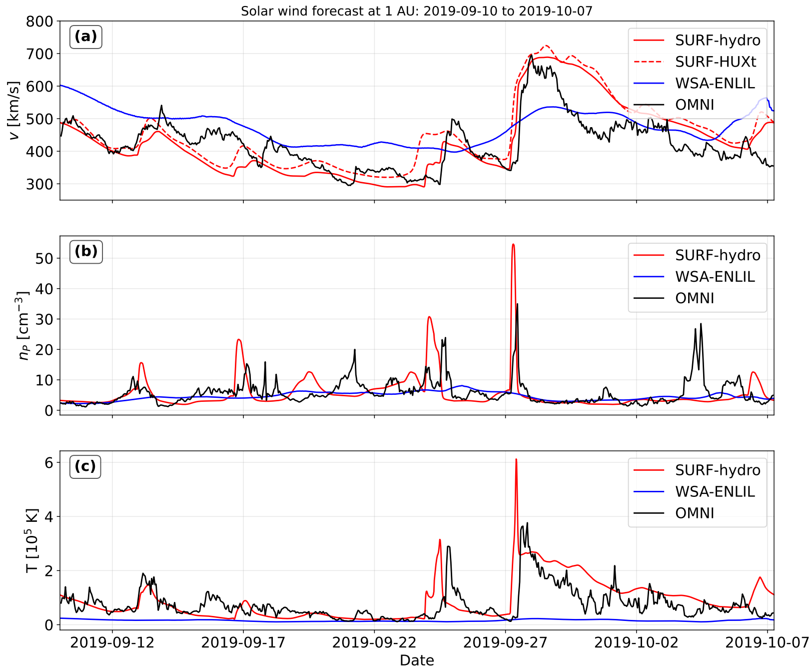

Figure 6. Solar wind hindcasts made on on 2019-9-10 for the next 27 days using SURF and the input conditions shown in Figure 5. Black lines show OMNI observations, dashed red lines show SURF-HUXt, and solid red lines show SURF-hydro. (a) Solar wind speed at 1 AU, (b) proton number density at 1 AU, and (c) proton temperature at 1 AU. For comparison, a series of 5-day forecasts from the Space Weather Prediction Center's operational WSA-Enlil forecasts are shown in blue.

by around 50%, which may be due to the 1D nature of the solution; deflection of flow away from the radial is not possible. It is also possible that by backmapping 1-AU values to 0.1 AU without undoing the effect of compression and rarefaction, then propagating out to 1 AU again, we have effectively double-counted the effect of stream interactions. This will be investigated further in study work.

However, it is important to put the scale of such errors into the wider context. The blue lines in Figure 6 show the Space Weather Prediction Center's (SWPC's) WSA-Enlil forecasts for the same period, obtained from the archive at the Community Coordinated Modelling Center (CCMC): https://iswa.ccmc.gsfc.nasa.gov/iswa_data_tree/model/heliosphere/wsa-enlil-cone/. The WSA-Enlil time series is constructed from four separate 5-day forecasts, so the forecast lead time is generally much lower that the SURF runs, which varies from 0 to 27 days over the interval shown. There is some evidence that the observed pattern in v is present in the WSA-Enlil solution, but the dynamic range is much reduced. There is almost no variation in n_P, while T is systematically an order of magnitude too low. Thus the SURF-hydro solution is providing much more accurate and useful information at this time. Of course, this is not a model issue; Enlil contains a more accurate representation of solar wind physics than SURF-hydro. But the solar wind solution at 1-AU is primarily determined by the model boundary conditions (Heinemann et al. 2025), not the model physics.

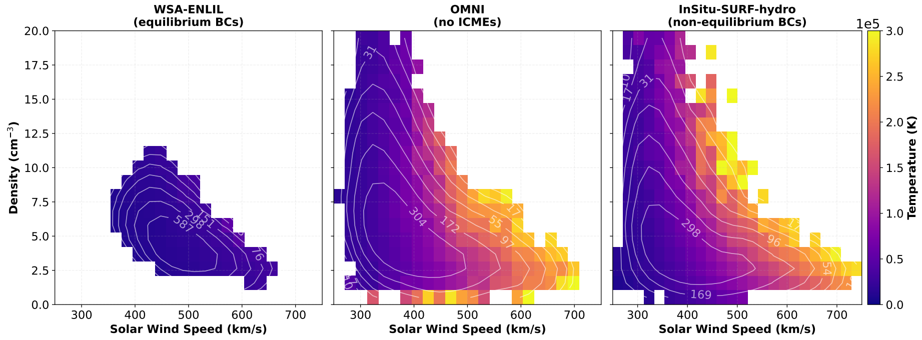

Figure 7. Solar wind conditions at 1 AU over the period 2019-2023 at 1-hour resolution for (left) WSA-Enlil, (centre) OMNI in situ observations, and (right) InSitu-SURF-hydro. Mean density is shown as a function of mean solar wind speed, with mean temperature indicated by the colour scale. Only points with at least 10 values are shown. The white contours show occurrence.

Figure 7 investigates the importance of correctly specifying the 0.1 AU plasma conditions in a more statistical fashion. The left hand panel shows 1 AU properties for WSA-Enlil solutions over the period 2019 through 2022. The centre panel shows the same analysis for the OMNI near-Earth solar wind observations, with periods identified as ICMEs by the Richardson and Cane (2010) catalogue removed. All data are averaged to 1-hour and then binned by 1-AU speed, density and temperature. Only parameter space containing at least 10 values is shown. There are a number of notable difference between WSA-Enlil and OMNI observations. Firstly, the range of speeds in WSA-Enlil is reduced compared to observations. This could be a result of under-dispersed speeds in the WSA solution. The lack of variability in speed could then lead to the observed lack of variability in density too, as there is reduced opportunity to form strong stream interaction regions. However, it is clear that the problem is more fundamental than that, as the WSA-Enlil temperatures are also far too low, even allowing for the reduced range of speeds produced by WSA-Enlil. This must be an issue with specifying the inner-boundary temperatures, suggesting that the equilibrium assumption employed in WSA-Enlil is limiting the accuracy of the resulting simulations.

The right-hand panel of Figure 7 shows the SURF-hydro solution, with initial speeds specified by backmapping the OMNI observed speeds. Thus the fact that SURF-hydro matches the range of speeds has been explicitly built into the reconstruction. What is important, however, is that the 0.1-AU densities and temperatures that are derived using the empirical relations in Table 2 then produce the correct range of densities and temperatures at 1 AU. It also shows broadly similar correlations to those observed, with fast wind being less dense and hotter than slow wind. We therefore suggest that WSA-Enlil (and other coupled corona-heliosphere models) could be significantly improved by employing similar empirical density and temperature relations, rather than the current equilibrium assumptions that are used.

It should be noted that these examples only consider one point in the heliosphere; near-Earth space. Thus the radial acceleration could still be systematically biassed. It is important to test against inner heliosphere observations, such as provided by *Parker Solar Probe* and *Solar Orbiter*. This will provide a much stronger constraint on the solar wind acceleration (e.g. Badman et al. 2025; Rivera et al. 2025, 2024).

6. Coronal mass ejections

The primary purpose of a space-weather modelling system is to represent the time-dependent flows associated with coronal mass ejections (CMEs). The standard operational forecasting approach is to characterise the kinematics of CMEs – speed, direction, angular width – by fitting a simple geometric model to white-light coronagraph observations (Zhao, Plunkett, and Liu 2002). A perturbation at the solar wind model inner boundary is then introduced with the same basic properties (Odstrcil, Riley, and Zhao 2004). This localised, transient perturbation can be a simple speed perturbation, or also involve changes to the solar wind density and temperature at 0.1 AU. There has not been a great deal of systematic research into the exactly how best these perturbations should be made. For example, the WSA-Enlil operational forecasting system inserts a perturbation with the coronagraph-estimated CME speed, a typical ambient solar wind temperature at 0.1 AU and four times the ambient solar wind density at 0.1 AU (Zheng et al. 2013; Taktakishvili et al. 2009). Uncertainty in the density and temperature of CMEs is not something that is currently considered in ensemble forecasting (Mays et al. 2015), primarily because the parameter space is large and multiple runs of 3DMHD models are computationally expensive. Very limited sensitivity analysis of these properties has been performed for a single event (Falkenberg et al. 2010), though this has primarily focussed on further increasing the CME perturbation to be hotter and denser than the ambient solar wind conditions at 0.1 AU. The rationale behind this is that the cone-model representation of CMEs neglects the strong internal magnetic field of CMEs, and thus increased thermal pressure is required to make up for the lack of internal magnetic pressure (see also Verbeke et al. 2022).

However, ICMEs at 1 AU are known to be very low proton density and proton temperature (Richardson and Cane 2010) compared to ambient solar wind conditions. To demonstrate this, Figure 8 shows a super-posed epoch analysis of all magnetic clouds in the Richardson and Cane (2010) catalogue with an average speed greater than 500 km s^{-1}. We choose the fast events so that the sheath region ahead of the ICME can be included too. The median sheath duration for these events is 7.1 hours, while the median ICME duration is 26 hours. All event time series are normalised to these durations. For this 'typical' event, the sheath region is characterised by fast, dense and hot solar wind. The ICME

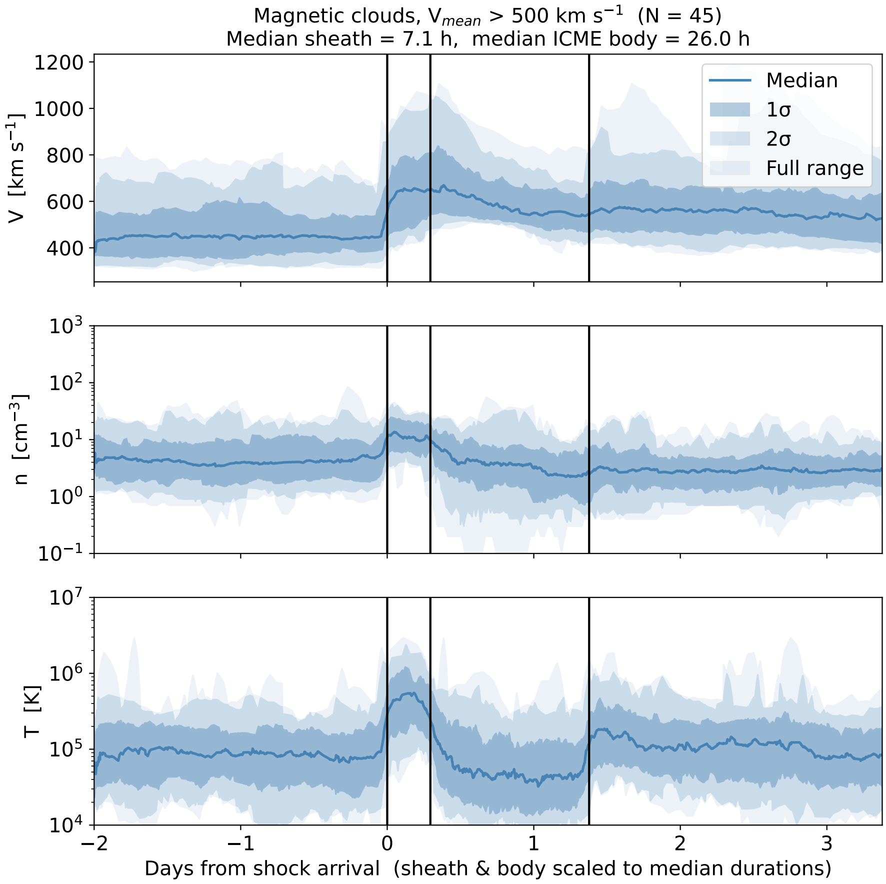

Figure 8. Super-posed epoch analysis of OMNI near-Earth solar wind data for all fast magnetic clouds between 1994 and 2025 (N = 45). Top: solar wind speed, middle: proton density, bottom: temperature. The median sheath duration for these events is 7.1 hours, and the median ICME body duration is 26 hours. All event durations have been normalised to these values to show typical variations. The vertical lines show the shock, ICME leading and trailing edges identified in the Richardson and Cane (2010) catalogue. The median value at each time is shown by the blue line, with lighter shades showing 1-σ, 2-σ and the full range of values.

body is characterised by a declining speed profile, and lower density and temperature than the surrounding ambient solar wind. This density and temperature reduction partly arises from adiabatic expansion during transit from 0.1 to 1 AU (Démoulin and Dasso 2009; Agarwal and Mishra 2024). But there is also evidence that CMEs close to the Sun have already undergone significant expansion and cooling (Romeo et al. 2023; Davies et al. 2024). Therefore it seems plausible that CMEs at 0.1 AU should be represented as low density and temperature perturbations at 0.1 AU, rather than the current over-dense perturbations used operationally.

SURF-hydro is well suited to investigating this problem, as parameter space can be rapidly sampled. The magnetic pressure is neglected, but as discussed above, even in 3DMHD models CMEs are generally treated as hydrodynamic structures. Thus the core physics of adiabatic expansion and cooling is still well represented. Indeed, 1D hydrodynamic models have strong heritage in successful interpretation of CME dynamics and expansion (e.g. Gosling et al. 1998).

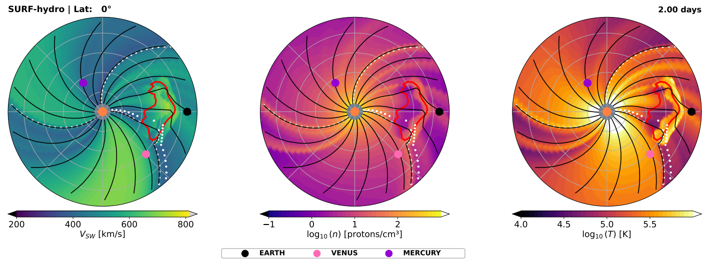

Figure 9. An example of the solar equatorial plane for a SURF-hydro solution for an arbitrary CME (directed at Earth, with angular width of 60° and speed 800 km s^{-1}) into a random WSA solar wind solution of v and B_R at 0.1 AU. Ambient density and temperature are determine from Table 2. The CME is represented by a v perturbation only; all other CME parameters are left at ambient values. Panels, from left to right, show snapshots of v, $\log_{10}(n_P)$ and $\log_{10}(T)$, respectively, 2 days after the CME launch. Red lines show the CME boundary, as determined by tracer particles. Black lines show streak lines of the flow, which approximate heliospheric magnetic field lines. White dots show the location of the heliospheric current sheet.

Figure 9 shows an example of typical CME solution in SURF-hydro. It uses a randomly-selected WSA solution to provide ambient solar wind v and B_R at 0.1 AU. v at 0.1 AU is reduced using the relations in the first two columns of Table 2. Density and temperature at 0.1 AU are generated by interpolating the remaining values in Table 2. These boundary conditions are used to generate a structured ambient solar wind that contains the heliospheric current sheet (HCS) within the slow wind, shown as the white dashed line in Figure 9. Additional heliospheric magnetic field (HMF) lines are traced as streak lines in the flow, and shown as black lines in all panels of Figure 9. Enhanced density and temperature fronts, aligned with the Parker spiral, are seen where fast wind meets slow (Pizzo 1991).

Into this ambient solar wind, a CME is inserted as a time-dependent speed perturbation at 0.1 AU, beginning at $t = 0$ and lasting for 12 hours (Owens et al. 2025). It has angular width of 60° and speed of 800 km s^{-1}. For this run, the density and temperature in the CME perturbation are left at ambient values, rather than inserting an over-dense blob. Tracer particles are inserted at 0.1 AU at the time that the CME leading and trailing edges pass the inner boundary. This results in the CME boundary shown as a red contour. It can be seen that there is hot, dense plasma ahead of the CME leading edge; this is the CME sheath region bounded by the shock front (Owens et al. 2005; Siscoe and Odstrcil 2008). This compression also propagates back into the CME body. The remainder of the CME body is cooler and less dense than the surrounding ambient wind, due to greater radial expansion of the ambient wind and hence greater adiabatic cooling.

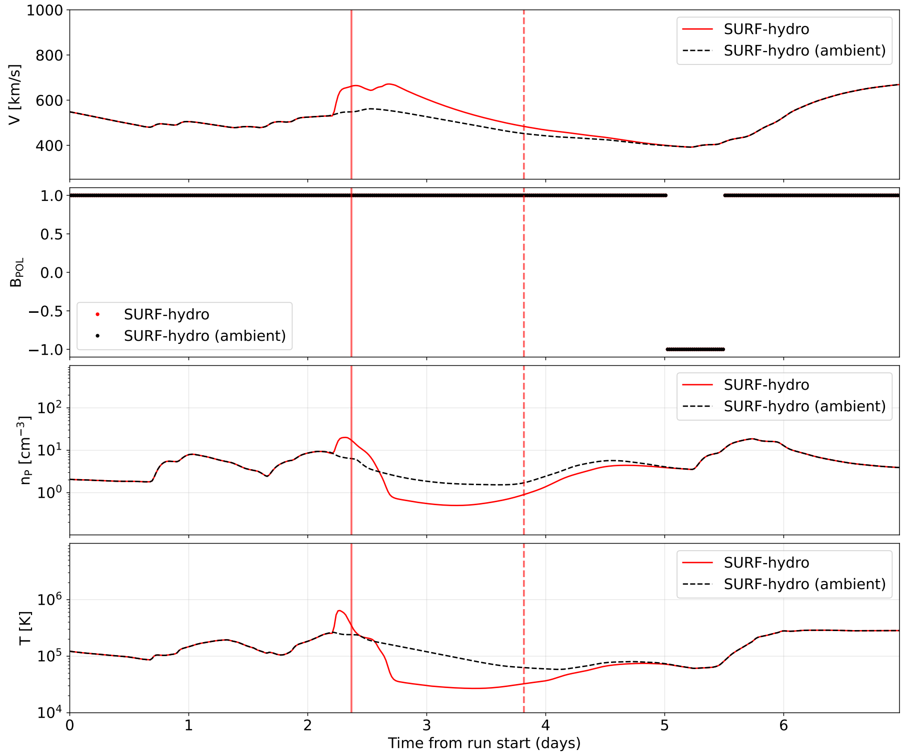

Figure 10. Time series of solar wind properties at Earth for the SURF-hydro solution shown in Figure 9. Red lines show the CME run, black dashed lines show the ambient solar wind run. Panels, from top to bottom, show v, the HMF polarity $\mathrm{B_{POL}}$, $\mathrm{n_P}$ and T. Solid red vertical lines show the time of the CME leading edge, while vertical red dashed lines show the CME trailing edge. These are determined by passive tracer particles in the flow.

Figure 10 shows the resulting time series at Earth, for both an ambient wind run of SURF-hydro (the black dashed lines) and with the CME present (the red solid lines). The vertical lines show the CME leading and trailing edges at Earth determined from the tracer particles. As seen from the snapshot in Figure 9, the interval of solar wind disturbed from the ambient values is greater in extent than the CME perturbation itself. The true time between the shock front and the CME leading edge is only a couple of hours, though if the CME leading edge were to be identified as the time when the density and temperature drops (as appears to be the case with the Richardson and Cane 2010, identifications), it would be misidentified as more than twice that. This longer estimate is in close agreement with the observed typical sheath ahead of an ICME shown in Figure 8. The main body of the CME is indeed characterised by lower density and temperature than in the ambient wind, along with the characteristic declining velocity profile, indicative of on-going expansion in the radial direction (i.e., beyond the spherical expansion of the ambient solar wind). The CME trailing edge is more difficult to identify from these plasma signatures alone, and the tracer particles show that the CME trailing edge occurs before the return to ambient values. Both the true and apparent CME durations are a little over 24 hours, again in approximate agreement with typical observations (Richardson and Cane 2010, and Figure 8). Thus SURF-hydro reproduces many of the observed features of CMEs at 1 AU.

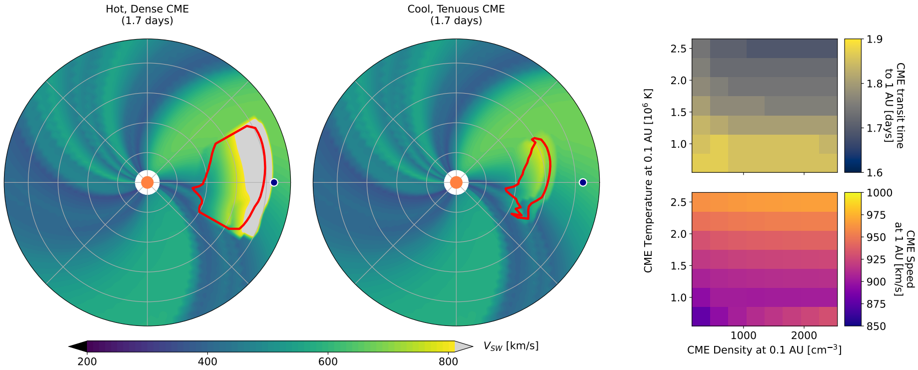

Figure 11. Left and centre: Two SURF-hydro runs after 1.7 days of the same ambient solar wind and with a CME with the same speed, direction and angular width, but different density and temperatures at 0.1 AU. Right: CME arrival time (top) and CME arrival speed (bottom) for a range of initial CME densities and temperatures.

Figure 11 shows a preliminary example of the kind of sensitivity analysis that is made more accessible by SURF-hydro. The two circular plots on the left-hand side show solar wind speed in the solar equatorial plane for very similar SURF runs; the same ambient solar wind, the same CME speed, direction and angular width, but different initial CME densities and temperatures at 0.1 AU. The right-hand plot uses 0.1 AU CME values of $n_P = 2000$ cm^{-3} and $T = 2.5 \times 10^6$ K, whereas the middle plot shows a CME with initial values of $n_P = 500$ cm^{-3} and $T = 1 \times 10^6$ K. There are significant differences in the resulting solar wind structures. The hot, dense CME arrives at Earth with a shorter transit time and a faster arrival speed, expands more to have greater radial extent and drives a stronger shock with a thicker sheath region.

The two-dimensional colourmaps on the right-hand side of Figure 11 investigate the effect of initial CME density and temperature on the resulting 1-AU transit time and 1-AU arrival speed for this specific ambient solar wind structure. In general, temperature decreases transit time and increases arrival speed, primarily owing to the increasing CME expansion. The relation with density is more complex. For higher temperatures, there is a weak negative correlation between transit time and density. This is less prevalent at lower temperature. The key takeaway from this plot is that – for this ambient solar wind structure and CME speed – the 1-AU transit time and arrival speed can vary by around 15-20%, depending on the initial CME parameters. Thus the initial density and temperature of CMEs merits more study, and ensemble forecasts should likely include perturbations to these highly uncertain parameters.

7. Summary and discussion

In this study, we outlined and tested a one-dimensional, compressible hydrodynamic model of the solar wind. This builds on the previous incompressible "HUXt" model. In order to take of advantage of the considerable toolset built up around the HUXt model, we have repackaged the general framework as "SURF"; Space-weather Utilities for Research and Forecasting. HUXt and the new "hydro" models are options within SURF. While hydro is not as computationally efficient as HUXt, it is still rapid enough for use in large ensembles and parametric studies, which can be used to inform and improve future 3DMHD modelling. For example, a single SURF-HUXt 1D run for 5 days – adequate to predict CME propagation to Earth – takes around 0.02 seconds on standard desktop CPU. The same run with SURF-hydro is around 0.1 seconds. This is still around $\times 10^4$ less computational resource than 3DMHD. And as SURF is all written in pure python, it is much simpler to set up and run on standard computer and to integrate within existing forecast and analysis frameworks.

An obvious advantage to solving for density and temperature – rather than just the solar wind flow speed as in HUXt – is that it is possible to use SURF-hydro output to construct synthetic heliospheric imager observations (Eyles et al. 2009; Howard and DeForest 2012; Odstrcil et al. 2020) with relatively low computational overhead. This potentially enables more rigorous data assimilation of heliospheric imager observations (Barnard et al. 2019, 2020, 2023) to improve solar wind reconstructions for both scientific and forecasting use.

Most coronal models provide radial speed and magnetic field as inner boundary conditions to solar wind models (e.g. Arge et al. 2003; Riley, Linker, and Mikic 2001), typically at 0.1 AU. Thus it is necessary to derive density and temperature from the provided speed. The standard method is to assume form form of equilibrium state at the 0.1 AU inner boundary surface; either constant mass, momentum or kinetic energy flux to determine density, then constant thermal pressure to determine temperature (Pomoell and Poedts 2018). SURF-hydro allows these assumptions to be rigorously tested with comparison to decades of observations. This will form the basis of a future study. Here, however, we have proposed a simple non-equilibrium method to determine inner boundary density and temperature. We use empirical relations derived using 30 years of near-Earth solar wind observations, then "backmap" relations (allowing for acceleration and expansion) to 0.1 AU. This non-equilibrium method is shown to produce sensible values at 1 AU and to perform favourably in comparison to state-of-the-art methods like WSA-Enlil (Zheng et al. 2013). Having established the effectiveness of this approach with SURF-hydro, this methodology could be easily applied to other solar wind models.

Properties of CMEs at 0.1 AU are even more uncertain than the ambient solar wind boundary conditions. CME speeds and directions are routinely characterised by coronagraphs (though with unknown, but likely very large uncertainties) and used to define time-dependent boundary condition perturbations in solar wind models (Zhao, Plunkett, and Liu 2002). For operational forecasting, the CME perturbation is assumed to have x4 ambient solar wind density and the same temperature as ambient wind (Taktakishvili et al. 2009). This is in contrast to observations at 1 AU that find CMEs to be cooler and more tenuous than the ambient solar wind, though this is expected to result from in-transit adiabatic expansion (Richardson and Cane 2010). Again, computational limitations mean that these modelling choices have not been systematically explored, something that SURF-hydro can potentially provide in a future study. Here, we show an example of how SURF-hydro can be employed to this issue. For a particular structured solar wind, CME density and temperature at 0.1 AU affects propagation time and arrival speed at Earth by around 15-20%. This will be different for different CMEs and ambient solar wind contexts, and could be assessed with suitable perturbations during ensemble forecasting.

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Author Contribution Owens led the development and writing. Barnard led the code verification, curation and packaging.

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Data Availability All data are packaged with the analysis code at the Github repository below. WSA-Enlil data were retrieved from https://iswa.ccmc.gsfc.nasa.gov/iswadatatree/model/heliosphere/wsa-enlil-cone/, ONMI data wer retrieved from https://omniweb.gsfc.nasa.gov/.

Materials Availability N/A

Code Availability Github and Zenodo links provided on acceptance.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

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